

Tree equivalence theorem contains a mistake

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Solution

A theorem presented in Piotr Chrzastowski-Wachtel, Boualem Benatallah, Rachid Hamadi, Milton O'Dell, Adi Susanto, Top-down Petri Net Based Approach to Dynamic Workflow Modelling, 2003 states the following:

Definition 2. *Two trees T_1 with root v_1 and T_2 with root v_2 are equivalent iff one of the following conditions holds*

- T_1 and T_2 are one-node trees and the types of the nodes is the same.
- If v_1 and v_2 are labeled by the sequential refinements, then the number of children of v_1 is the same as the number of children of v_2 , and the corresponding children are equivalent.
- If v_1 and v_2 are labeled by OR-split, AND-split, or Loop-split, then there is a bijection between equivalent children of v_1 and v_2 .

It turns out that whenever different orders of refinements lead to the same net, then the contracted trees are identical for both of them.

Theorem 3. *If the WF net $\mathcal{N}$ resulted from a sequence of considered refinements then, no matter in which order they were made, the resulting refinement tree is uniquely determined.*

The theorem contains a mistake. Observe that a sequence: Loop from place v ; Loop from place v

Is the same as: Loop from a place v , "or" transformation from the loop transition. The trees are obviously can't be equivalent according to the equivalence definition, which contradicts the theorem.

To fix the theorem, during the "compression" of a tree we can force the following changes: When a node o of type "or" is a children of a node l of type "loop", we bring all children of o to be children of l , then remove o .

Since no other sequence of transformations is equivalent to some different one, this fixes the theorem.